

	Hanoi University of Science and Technology Faculty of Mathematics and Informatics	
	Subject: Calculus 1	ID Course: MI1111E
	Academic year: 2026.1	Training program: Bachelor
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1.1 Week 1

Question 1. Find the domain of the functions

- | | | |
|--|---------------------------------------|--|
| (a) $y = \sqrt{2 \operatorname{arccot} x - \pi}$ | (f) $y = \frac{\sqrt{x}}{\sin \pi x}$ | (k) $y = \arcsin(1 + 2x)$ (CK20221) |
| (b) $y = \sqrt{\pi - 2 \operatorname{arccot} x}$ | (g) $y = \frac{\ln x}{\sin \pi x}$ | (l) $y = \arcsin(1 + 3x)$ (CK20221) |
| (c) $y = \sqrt{\pi - 3 \operatorname{arccot} x}$ | (h) $y = \arccos(\sin x)$ | (m) $y = \sin \sqrt{x - 1}$ (CK20221) |
| (d) $y = \sqrt{\pi - 2 \arctan x}$ | (i) $y = \arcsin(\sin x)$ | (n) $y = \cos \sqrt{x + 1}$ (CK20221) |
| (e) $y = \arcsin \frac{2x}{1 + x}$ | (j) $y = \sin(\arcsin x)$ | (o) $y = \frac{1}{\sqrt{\frac{\pi}{2} - \arccos x}}$ (GK20221) |

Question 2. Find the range of the functions

- | | |
|--|---|
| (a) $y = \log(1 - 2 \cos x)$ | (e) $y = \arctan \sqrt{1 - x^2}$ (CK20221). |
| (b) $y = \arcsin \left(\log \frac{x}{10} \right)$ | (f) $y = \operatorname{arccot} \sqrt{1 - x^2}$ (CK20221). |
| (c) $y = \operatorname{arccot}(\sin x)$ | (g) $y = \frac{\pi}{\operatorname{arccot} x}$ (CK20221). |
| (d) $y = \arctan(e^x)$ | |

Question 3. Determine whether the following functions are even or odd.

- | | |
|---|--|
| (a) $f(x) = \sin \left(\frac{\sqrt{1+x} + \sqrt{1-x}}{x} \right)$ (GK20221). | (c) $y = \sin x + \cos^2 x$ (CK20141). |
| (b) $y = \ln(\sqrt{4x^2 + 1} - 2x)$. | (d) $y = \ln(\sqrt{x^2 + 1} - x)$ (CK20201). |

Question 4. Prove the equalities

- | | |
|--|--|
| (a) $\sinh(-x) = -\sinh x$ | (d) $\sinh 2x = 2 \sinh x \cosh x$ |
| (b) $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$ | (e) $\cosh^2 x - \sinh^2 x = 1$ |
| (c) $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$ | (f) $\cosh 2x = \cosh^2 x + \sinh^2 x$ |

Question 5. Find the function $f(x)$, where

$$(a) \ f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$$

$$(b) \ f\left(\frac{x}{1+x}\right) = x^2$$

Question 6. Find the inverse function of

$$(a) \ y = 2 \arcsin x$$

$$(b) \ y = \frac{1-x}{1+x}$$

$$(c) \ y = \frac{1}{2}(e^x - e^{-x})$$

Question 7. (a) Let $f(x) = x + e^x$ and $g(x) = \frac{x+1}{2x-1}$. Find $(f^{-1} \circ g^{-1} \circ g^{-1} \circ f)(0)$?

(b) Let $f(x) = 2x + \ln(x)$ and $g(x) = \frac{1}{7}(15 - x^3)$. Find $(f^{-1} \circ g^{-1} \circ g^{-1} \circ f)(1)$.

Question 8. Given two functions $f(x) = 1 + \frac{\pi x}{2}$ and $g(x) = x + \arcsin x$. Evaluate $(f \circ g^{-1} \circ f)(1)$.

$$A. \ 1 + \frac{\pi}{2}$$

$$B. \ 1$$

$$C. \ \pi$$

$$D. \ \frac{\pi}{2}$$

1.2 Week 2

Question 9. Determine whether the sequence converges or diverges. If it converges, find the limit

$$(a) \ a_n = n - \sqrt{n^2 - n}$$

$$(c) \ a_n = \frac{\sin^2 n - \cos^3 n}{n}$$

$$(b) \ a_n = \frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{(n-1)n}$$

$$(d) \ a_n = \frac{\sqrt{n} \cos n}{n+1}$$

Question 10. Find the limit

$$(a) \ \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+n}} \right) \quad (b) \ \lim_{n \rightarrow \infty} \frac{n^2}{a^n} \text{ (with } a > 1 \text{)}$$

Question 11. Prove that the following sequences converge.

$$(a) \ x_n = \frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2}$$

$$(b) \ x_n = \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$$

Question 12. Find $\lim_{n \rightarrow \infty} x_n$, where $x_1 = \sqrt{2}$, $x_{n+1} = \sqrt{2 + x_n}$ $n \geq 1$.

Question 13 (✓). Given $\lim_{n \rightarrow \infty} x_n = a$. Find the limit

$$\lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \dots + x_n}{n}.$$

Question 14 (✓). Given $x_n > 0$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} x_n = a$. Compute the limit $\lim_{n \rightarrow \infty} \sqrt[n]{x_1 x_2 \dots x_n}$.

Question 15. Prove that the following sequences diverge.

$$(a) \ x_n = (-1)^n \left(3 + \frac{3}{n} \right),$$

$$(b) \ x_n = 1 + \frac{n}{n+1} \cos \frac{n\pi}{2}.$$

$$(c) \ x_n = \sin n$$

Question 16. Prove that if $\lim_{n \rightarrow \infty} x_n$ exists and equals to L , then

- (a) $\lim_{n \rightarrow \infty} x_{n+2} = L.$ (b) $\lim_{n \rightarrow \infty} x_{n-1} = L.$ (c) $\lim_{n \rightarrow \infty} x_{2n} = L.$ (d) $\lim_{n \rightarrow \infty} x_{2n+1} = L.$

Question 17. Find the limit

- (a) $\lim_{x \rightarrow 0} \left(\frac{1}{x} \sqrt{1+x} - \frac{1}{x} \right)$ (d) $\lim_{x \rightarrow 0} \frac{\sqrt[m]{1+\alpha x} - \sqrt[n]{1+\beta x}}{x}, \quad (m, n \in \mathbb{N}^*)$
- (b) $\lim_{x \rightarrow +\infty} (\sqrt[3]{x^3 + x^2 - 1} - x)$ (e) $\lim_{x \rightarrow +\infty} x (\sqrt{x^2 + 2x} - 2\sqrt{x^2 + x} + x)$
- (c) $\lim_{x \rightarrow 1} \frac{x^{100} - 2x + 1}{x^{50} - 2x + 1}$ (f) $\lim_{x \rightarrow 0} \frac{\sqrt{1+4x} - 1}{\ln(1+3\sin x)}$

Question 18. Find the limit

- (a) $\lim_{x \rightarrow 0^+} \frac{\ln(x + \arccos^3 x) - \ln x}{x^2}$ (c) $\lim_{x \rightarrow 0} \frac{\sqrt{\cos x} - \sqrt[3]{\cos x}}{\sin^2 x}$
- (b) $\lim_{x \rightarrow +\infty} (\sin \sqrt{x+1} - \sin \sqrt{x})$ (d) $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{1 - \cos x}$

Question 19. Find the limit

- (a) $\lim_{x \rightarrow \infty} \left(\frac{x^2 - 1}{x^2 + 1} \right)^{\frac{x-1}{x+1}}$ (e) $\lim_{x \rightarrow \infty} \left(\sin \frac{1}{x} + \cos \frac{1}{x} \right)^x$
- (b) $\lim_{x \rightarrow 0^+} (\cos \sqrt{x})^{\frac{1}{x}}$ (f) $\lim_{x \rightarrow 1} (1 + \sin \pi x)^{\cot \pi x}$
- (c) $\lim_{x \rightarrow \infty} [\sin(\ln(x+1)) - \sin(\ln x)]$
- (d) $\lim_{n \rightarrow \infty} n^2 (\sqrt[n]{x} - \sqrt[n+1]{x}), x > 0.$ (g) $\lim_{x \rightarrow 0} [\ln(e+2x)]^{\frac{1}{\sin x}}$

Question 20. As $x \rightarrow 0^+$, are the following pairs of infinitesimals equivalent?

$$\alpha(x) = \sqrt{x + \sqrt{x}} \quad \text{and} \quad \beta(x) = e^{\sin x} - \cos x.$$

Question 21. Compare two following infinitesimal functions:

- (a) $\alpha(x) = \sqrt{x + \sqrt{x}}$ and $\beta(x) = e^{\sin x} - \cos x$, as $x \rightarrow 0^+$
- (b) $\alpha(x) = \sqrt[3]{x} - \sqrt{x}$ and $\beta(x) = \cos x - 1$, as $x \rightarrow 0^+$
- (c) $\alpha(x) = x^3 + \sin^2 x$ and $\beta(x) = \ln(1 + 2 \arctan(x^2))$, as $x \rightarrow 0$

Question 22. Find all values of a such that f is continuous at $x = 0$:

- (a) $f(x) = \begin{cases} \frac{1 - \cos x}{x^2}, & \text{if } x \neq 0, \\ a, & \text{if } x = 0. \end{cases}$ (b) $g(x) = \begin{cases} ax^2 + bx + 1, & \text{if } x \geq 0, \\ a \cos x + b \sin x, & \text{if } x < 0. \end{cases}$

Question 23. What types of discontinuity does the following function have at $x = 0$?

$$(a) \ y = \frac{8}{1 - 2^{\cot x}}$$

$$(c) \ y = \frac{\sin \frac{1}{x}}{e^{\frac{1}{x}} + 1}$$

$$(b) \ y = \frac{1}{x} \arcsin x$$

$$(d) \ y = \frac{e^{ax} - e^{bx}}{x}, \ (a \neq b)$$

Question 24. (GK20221) Find and classify the discontinuities of $y = \frac{x}{\sin x}$.

Question 25. (GK20221) Compare the infinitesimals as $x \rightarrow 0^+$:

$$\alpha(x) = \frac{x(\sqrt{x+1} - \sqrt{x})}{2} \text{ and } \beta(x) = \ln \left(\frac{2x+1}{x+1} \right).$$

Question 26. (GK20221) As $x \rightarrow +\infty$, is the function $f(x) = x \sin x$ bounded? Is it infinitely large? Explain.

Question 27. Find $\lim_{x \rightarrow 0^+} \left(\frac{\sin \sqrt{x}}{2} \right)^x$.

A. 0

B. $\frac{e}{2}$

C. $\frac{1}{2}$

D. 1

Question 28. Find $a, \alpha \in \mathbb{R}$ such that the infinitesimal $\alpha(x) = \sqrt[3]{x} + \sqrt{x}$ is equivalent to ax^α as $x \rightarrow 0^+$:

A. $a = 1$ and $\alpha = \frac{1}{2}$.

B. $a = \frac{1}{3}$ and $\alpha = \frac{1}{3}$

C. $a = 1$ and $\alpha = \frac{1}{3}$

D. $a = \frac{1}{2}$ and $\alpha = \frac{1}{3}$

Question 29. Compute the limit $\lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$.

A. 1

B. $\frac{\pi}{3}$

C. $\frac{\pi}{2}$

D. 0

Question 30. How many infinitesimals among the following have a lower order than the infinitesimal $\alpha(x) = \tan x$ as $x \rightarrow 0^+$:

$$\alpha_1(x) = \sqrt[3]{x}, \quad \alpha_2(x) = \sqrt{x} + x^2, \quad \alpha_3(x) = \sqrt[4]{\sin x}, \quad \alpha_4(x) = x$$

A. 4

B. 1

C. 2

D. 3

1.3 Week 3

Question 31. Find the derivative of the function

$$f(x) = \begin{cases} 1 - x, & \text{if } x < 1, \\ (1 - x)(2 - x), & \text{if } 1 \leq x \leq 2, \\ x - 2, & \text{if } x > 2. \end{cases}$$

Question 32. Suppose f is a function that satisfies the equation $f(x + y) = f(x) + f(y) + x^2y + xy^2$, for all x, y , and $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$. Find $f'(x)$.

Question 33. Determine the constants a, b, c and d such that f is differentiable on $\mathbb{R}$.

$$(a) \quad f(x) = \begin{cases} 4x & \text{if } x \leq 0 \\ ax^2 + bx + c & \text{if } 0 < x < 1 \\ 3 - 2x & \text{if } x \geq 1. \end{cases}$$

$$(c) \quad f(x) = \begin{cases} ax + b & \text{if } x \leq 0 \\ ax^2 + c & \text{if } 0 < x \leq 2 \\ \frac{dx^2 + 1}{x} & \text{if } x \geq 2. \end{cases}$$

$$(b) \quad f(x) = \begin{cases} ax + b & \text{if } x \leq 0 \\ cx^2 + dx & \text{if } 0 < x \leq 1 \\ 1 - \frac{1}{x} & \text{if } x > 1. \end{cases}$$

Question 34. Find the derivatives of the following functions:

$$(a) \quad y = x^x, \quad 0 < x \neq 1.$$

$$(c) \quad y = \log_x 2, \quad 0 < x \neq 1.$$

$$(b) \quad y = (1 + \sin x)^x.$$

$$(d) \quad y = \log_x(\cos x), \quad x \in (0, \frac{\pi}{2}) \setminus \{1\}.$$

Question 35. Prove that the following equations have a unique solution.

$$(a) \quad x^{13} + 7x^3 - 5 = 0$$

$$(c) \quad 4^x + 5^x = 3^x$$

$$(b) \quad 3^x + 4^x = 5^x$$

Question 36. Solve the equation $3^x + 5^x = 4^x$.

Question 37. Find $f'(x)$ where $\frac{d}{dx}[f(2017x)] = x^2$.

Question 38. Prove that $\lim_{x \rightarrow 0} \left(x^n \sin \frac{1}{x} \right) = \begin{cases} 0 & \text{if } n > 0, \\ \infty & \text{if } n \leq 0. \end{cases}$

Question 39. Under what conditions is the function

$$f(x) = \begin{cases} x^n \sin \frac{1}{x}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0 \end{cases} \quad (n \in \mathbb{Z})$$

(a) continuous at $x = 0$?

(c) continuously differentiable at $x = 0$?

(b) differentiable at $x = 0$?

Question 40. Prove that the function $f(x) = |x - a|\varphi(x)$, where $\varphi(x)$ is a continuous function and $\varphi(a) \neq 0$, is not differentiable at $x = a$.

Question 41. Find the differential of each function

$$(a) \quad y = \frac{1}{a} \arctan \frac{x}{a}, \quad (a \neq 0)$$

$$(c) \quad y = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right|, \quad (a \neq 0)$$

$$(b) \quad y = \arcsin \frac{x}{a}, \quad (a \neq 0)$$

$$(d) \quad y = \ln |x + \sqrt{x^2 + a}|.$$

Question 42. Find

$$(a) \frac{d}{d(x^2)} \left(\frac{\sin x}{x} \right)$$

$$(b) \frac{d(\sin x)}{d(\cos x)}$$

$$(c) \frac{d}{d(x^3)} (x^3 - 2x^6 - x^9).$$

Question 43. (GK20211) How many natural numbers n make the function

$$f(x) = |x|^{5-n} \sin x$$

differentiable at $x = 0$?

A. 6

B. 4

C. 5

D. 7

Question 44. The equation of the tangent line to the curve given by $\begin{cases} x = 2t^2 + 1 \\ y = t^3 - 3 \end{cases}$ at $M(9, 5)$ is

A. $3x - 2y - 17 = 0$

B. $x + y - 14 = 0$

C. $2x - y - 13 = 0$

D. $x - y - 4 = 0$

Question 45. Consider the function $f(x) = \frac{e^{2x}}{x}$. Which of the following statements is correct?

A. f has a local minimum at $x = 2$.

C. f has a local maximum at $x = \frac{1}{2}$.

B. f has a local minimum at $x = \frac{1}{2}$.

D. f has a local maximum at $x = 2$.

Question 46. Is $x = 1$ a point of which type of discontinuity for the function $y = \arctan\left(\frac{1}{x-1}\right)$?

A. type 2

C. removable

B. not a discontinuity

D. type 1

Question 47 (✓). How many functions $f(x)$ are differentiable on $\mathbb{R}$ and satisfy

$$(f(x) - x^4)(f(x) - 2x^2 + 1) = 0, \forall x \in \mathbb{R}.$$

A. 2

B. 4

C. 6

D. 8

Question 48. What angle do the tangents to the parabola $y^2 = 4ax$ at the point with $x = a$ make with each other?

A. 0°

B. 30°

C. 45°

D. 90°

E. 120°

Question 49. Given $f(x) = \ln\left(\frac{2024x}{x+1}\right)$. Compute $S = f'(1) + f'(2) + f'(3) + \cdots + f'(2017)$.

A. $S = \frac{4035}{2018}$.

B. $S = \frac{2017}{2018}$.

C. $S = \frac{2016}{2017}$.

D. $S = 2017$.

Question 50. Find all values of m for which the function $y = \log_{2018}(2018^x - x - \frac{x^2}{2} - m)$ is defined for all x in $[0; +\infty)$.

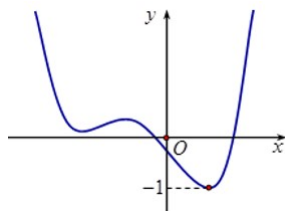
A. $m > 9$

B. $0 < m < 1$

C. $m < 1$

D. $m < 2$

Question 51. Given the function $y = f(x)$ with the graph shown below



Find the number of extreme points of the function $y = 2^{f(x)} - 3^{f(x)}$.

- A. 6. B. 5. C. 4. D. 3.

Question 52. Write the equation of the tangent line to the graph of the function $y = f(x) = x^x$ at the point where the x -coordinate is 2.

- A. $y = 4x - 4$. C. $y = 4(1 + \ln 2)x - 8 \ln 2 - 4$.
 B. $y = (4 \ln 2)x - 8 \ln 2 + 4$. D. $y = 2x$.

Question 53. Find all positive real values of the parameter a ($a > 0$) that satisfy

$$(2^a + \frac{1}{2^a})^{2021} \leq (2^{2021} + \frac{1}{2^{2021}})^a.$$

- A. $0 < a < 1$. B. $1 < a < 2021$. C. $a \geq 2021$. D. $0 < a \leq 2021$.

Question 54. Is the statement "*If the function $f(x)$ is differentiable at 0, then $f'(x)$ is continuous at 0*" true? Explain your answer.

1.4 Week 4

Question 55. Find $dy(-\frac{1}{2})$, where $y = \arccos \sqrt{1 - x^2}$.

- A. $\frac{2}{\sqrt{3}}dx$ B. $-\frac{2}{\sqrt{3}}dx$ C. $\frac{1}{\sqrt{3}}dx$ D. $-\frac{1}{\sqrt{3}}dx$

Question 56. Find the higher derivatives

- (a) $y = \frac{x^2}{1 - x}$, find $y^{(2020)}$ (d) $y = x^2 \sin x$, find $y^{(2020)}$
 (b) $y = \frac{1 + x}{\sqrt{1 - x}}$, find $y^{(2020)}$ (e) $y = e^{x^2}$, find $y^{(2020)}(0)$
 (c) $y = \ln(2x - x^2)$, find $y^{(2021)}$ (f) $y = x \ln(1 + 2x)$, find $y^{(2020)}(0)$

Question 57. Find the n -th derivative of the functions

- (a) $y = \frac{x}{x^2 - 1}$ (c) $y = \frac{x}{\sqrt[3]{1 + x}}$ (e) $y = \sin^4 x + \cos^4 x$
 (b) $y = \frac{1}{x^2 - 3x + 2}$ (d) $y = e^{ax} \sin(bx + c)$ (f) $y = x^{n-1} e^{\frac{1}{x}}$

Question 58. Given $y = \ln(1 - (x + 1)^{40})$. Find $y^{(2021)}(-1)$

Question 59. Find $y^{(2020)}(-1)$, where $y = \ln(1 - (x + 1)^4)$.

A. $\frac{-2020!}{40}$

B. $\frac{-2020!}{505}$

C. $\frac{2020!}{505}$

D. 0

Question 60. Find the higher-order differential of each function

(a) $y = (2x + 1) \sin x$. Find $d^{2020}y(0)$

(c) $y = x^9 \ln x$. Find $d^{2020}y(1)$

(b) $y = e^x \cos x$. Find $d^{2020}y(0)$

(d) $y = x^2 e^{ax}$. Find $d^{2020}y(0)$

Question 61. Prove that for each $\forall a, b, c \in \mathbb{R}$, the equation

$$a \cos x + b \cos 2x + c \cos 3x = 0$$

has at least one root in the interval $(0, \pi)$.

Question 62. Prove the equation $x^n + px + q = 0$, where n is a positive integer and $n \geq 2$, could not have at most 3 real roots if n is an odd number.

Question 63. Given three real numbers a, b, c such that $a + b + c = 0$. Prove that the equation $8ax^7 + 3bx^2 + c = 0$ has at least one root in $(0, 1)$.

Question 64. Assume that $a_0, a_1, \dots, a_n \in \mathbb{R}$ such that

$$\frac{a_0}{n+1} + \frac{a_1}{n} + \dots + \frac{a_{n-1}}{2} + a_n = 0.$$

Prove that the polynomial $P(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$ has at least one root in $(0, 1)$.

Question 65 (✓). The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice differentiable and satisfies $f(0) = 2$, $f'(0) = -2$ and $f(1) = 1$. Prove that there is $c \in (0, 1)$ such that $f(c) \cdot f'(c) + f''(c) = 0$.

Question 66. Given a continuous function $f : [0, 1] \rightarrow [0, 1]$. Prove that there is $c \in [0, 1]$ such that $f(c) = c$.

Question 67. Let f be a continuous on $[0, 1]$, differentiable on $(0, 1)$ and $f(0) = f(1) = 0$. Prove that there exists $c \in (0, 1)$ such that $f(c) + f'(c) = 0$.

Question 68. Given a continuous function $f : [0, 1] \rightarrow [0, 1]$ satisfying $f(0) = 0$ and $f(1) = 1$. Prove that there exists $c \in (0, 1)$ such that $f(c) = 1 - c$.

Question 69. Solve the equation $6^x + 2^x = 5^x + 3^x$.

Question 70. Given $f(x) = x(x-1)(x-2) \cdots (x-2022)$. Determine how many real solutions the equation $f'(x) = 0$ has?

Question 71. For $x > 0$, prove that:

(a) $2x \arctan x \geq \ln(1 + x^2)$ for all $x \in \mathbb{R}$

(c) $\cos x \leq 1 - \frac{x^2}{2} + \frac{x^4}{24}, \forall x \in \left(0, \frac{\pi}{2}\right)$

(b) $x - \frac{x^2}{2} \leq \ln(1 + x) \leq x$ for all $x \geq 0$

(d) $e^x \leq 1 + xe^x$, for all $x \in \mathbb{R}$.

(e) $xe^{\frac{x}{2}} < e^x - 1$

(h) $(\frac{x+1}{2})^{x+1} \leq x^x$

(f) $e^x - 1 - x < x^2 e^x$

(i) $(e+x)^{e-x} > (e-x)^{e+x}$, for $x \in (0, e)$.

(g) $e^x < (1+x)^{1+x}$

(j) $e^{x-1} + \ln x - 2x + 1 > 0, \forall x > 1$.

Question 72. Find the critical points of the following functions:

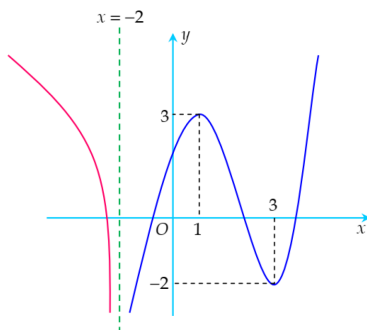
(a) $y = \frac{3x^2 + 4x + 4}{x^2 + x + 1}$

(c) $y = \sqrt[3]{(1-x)(x-2)^2}$

(b) $y = x - \ln(1+x)$

(d) $y = x^{\frac{2}{3}} + (x-2)^{\frac{2}{3}}$

Question 73. Given the function $f(x)$ is differentiable and its graph is as shown in the figure. Calculate the sum of all integer values of the parameter $m \in [-20, 20]$ such that the function $y = f(|x| + m)$ has 5 critical points.



Question 74. How many positive integer values of m make the function $y = |x^3 - 3x + m|$ have 5 critical points?

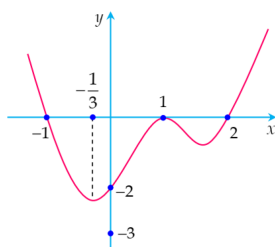
A. 1

B. 2

C. 3

D. 4

Question 75. Given a function $f(x)$, which is defined and differentiable on $\mathbb{R}$. The graph of f is provided below. Let $g(x) = f(f(x) - 1)$. Find the number of solutions to the equation $g'(x) = 0$.



A. 7

B. 8

C. 9

D. 6

1.5 Week 5

Question 76. Examine the differentiability of the following functions:

$$(a) f(x) = \begin{cases} x^2 e^{-x^2} & \text{if } |x| \leq 1 \\ \frac{1}{e} & \text{if } |x| > 1 \end{cases} \quad (b) f(x) = \begin{cases} \arctan \frac{1}{|x|} & \text{if } x \neq 0 \\ \frac{\pi}{2} & \text{if } x = 0. \end{cases}$$

Question 77. Find $f'(0)$, where $f(x) = \begin{cases} \frac{1 - \sqrt{1 + x^4} \cos(x^2)}{x^4 \ln(1 + 2x^3)} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

A. $-\frac{1}{3}$ B. $-\frac{1}{6}$ C. 1 D. $\frac{1}{2}$ E. $\frac{1}{6}$

Question 78. Explain why the Cauchy formula $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$ cannot apply for the functions $f(x) = x^2$ and $g(x) = x^3$, $-1 \leq x \leq 1$.

Question 79. Prove the following inequalities

(a) $|\sin x - \sin y| \leq |x - y|$ (c) $\frac{b - a}{1 + b^2} < \arctan b - \arctan a < \frac{b - a}{1 + a^2}, 0 < a < b$
(b) $\frac{a - b}{a} < \ln \frac{a}{b} < \frac{a - b}{b}, 0 < b < a.$

Question 80. Suppose that f is continuous on $[a, b]$, $a > 0$, and differentiable on (a, b) . Prove that if $\frac{f(a)}{a} = \frac{f(b)}{b}$, then there is $x_0 \in (a, b)$ such that $x_0 f'(x_0) = f(x_0)$.

Question 81 (✓). Suppose that f is continuous on $[a, b]$ and differentiable on (a, b) . Prove that if $f^2(b) - f^2(a) = b^2 - a^2$, the equation $f'(x)f(x) = x$ has at least one root in (a, b) .

Question 82. Find the limits:

(a) $\lim_{x \rightarrow +\infty} \left(\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x} \right)$ (e) $\lim_{x \rightarrow 1} \tan \frac{\pi x}{2} \ln(2 - x)$ (j) $\lim_{x \rightarrow +\infty} (x^3 + 3^x)^{\tan \frac{1}{x}}$
(b) $\lim_{x \rightarrow 1} \left(\frac{x}{x - 1} - \frac{1}{\ln x} \right)$ (f) $\lim_{x \rightarrow 0} (1 - a \tan^2 x)^{\frac{1}{x \sin x}}, (a \in \mathbb{R})$ (k) $\lim_{x \rightarrow 0} \frac{x^3}{x - \sin x}$
(c) $\lim_{x \rightarrow \infty} \frac{e^{\frac{1}{x}} - \cos \frac{1}{x}}{1 - \sqrt{1 - \frac{1}{x^2}}}$ (g) $\lim_{x \rightarrow 1^-} \frac{\tan \frac{\pi x}{2}}{\ln(1 - x)}$ (l) $\lim_{x \rightarrow 0} \frac{\sin x - x}{x(1 - \cos x)}$
(d) $\lim_{x \rightarrow 0} \frac{e^x \sin x - x(1 + x)}{x^3}$ (h) $\lim_{x \rightarrow 0} (1 - \cos x)^{\tan x}$ (m) $\lim_{x \rightarrow \infty} \left[\tan \left(\frac{\pi x}{2x+1} \right) \right]^{\frac{1}{x}}$
(i) $\lim_{x \rightarrow -\infty} (x^2 + 2^x)^{\frac{1}{x}}$ (n) $\lim_{x \rightarrow 0} \left(1 + x e^{-\frac{1}{x^2}} \sin \frac{1}{x^4} \right)^{\frac{1}{x^2}}$

Question 83. Find the limits

(a) $\lim_{x \rightarrow 0} \left(1 + e^{-\frac{1}{x^2}} \arctan \frac{1}{x^2} + x e^{-\frac{1}{x^2}} \sin \frac{1}{x^4} \right)^{\frac{1}{x^2}}$ (c) $\lim_{x \rightarrow 0^+} \left(2 \sin \sqrt{x} + \sqrt{x} \sin \frac{1}{x} \right)^x$
(b) $\lim_{x \rightarrow 0} \frac{\sin(\ln(x + 1)) - \ln(1 + \sin x)}{x^4}.$ (d) $\lim_{x \rightarrow 0} \frac{\sqrt{x + 1} - \sqrt{x + \cos x} \sqrt{\cos 2x}}{x^2}$

Question 84. Is there the function f such that $f(0) = -1$, $f(2) = 4$ and $f'(x) \leq 2$ for all x ?

Question 85. Determine the values a and b such that the following expression has a finite limit as $x \rightarrow 0$

$$f(x) = \frac{1}{\sin^3 x} - \frac{1}{x^3} - \frac{a}{x^2} - \frac{b}{x}.$$

Question 86. Let f be a real differentiable function on $[a, b]$ and there exists $f''(x)$ on (a, b) . Prove that for all $x \in (a, b)$ one can find at least $c \in (a, b)$ such that

$$f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a) = \frac{(x - a)(x - b)}{2} f''(c)$$

Question 87. Given $y = \begin{cases} \frac{1 - \sqrt{1 + x^4} \cos(x^2)}{x^4 \ln(1 + 2x^3)} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$. Find $y'(0)$.

Question 88 (✓). Let a_{2021} be the coefficient of the term of degree 2021 in the Maclaurin expansion of $f(x) = \cos(\sin(x))$. Calculate a_{2021} .

- A. 1 B. 2021! C. 0 D. 2021

1.6 Week 6

Question 89. Find all values of m for which the function $y = 8^{\cot x} + (m - 3) \cdot 2^{\cot x} + 3m - 2$ is increasing on $(\frac{\pi}{4}, \pi)$.

- A. $-9 \leq m < 3$ B. $m \leq 3$ C. $m \leq -9$ D. $m < -9$.

Question 90. Given the function $y = \left(\frac{2020}{2021}\right)^{-e^{5x} + (m+3)e^x + 2}$. It is known that for all $m \leq a \cdot e^b + c$ (where $a, b, c \in \mathbb{Z}$) the given function is increasing on the interval $(2; 5)$, find $S = \log_5 a^{c^b}$.

- A. $S = \frac{1}{125}$. C. $S = 6165$.
B. $S = 390625$. D. $S = 6561$.

Question 91. Investigate the monotonicity of the function.

- (a) $y = x^4 - 2x^3 + 2x - 1$ (c) $y = x + |\sin 2x|, x \in [0, \pi]$
(b) $y = 3 \arctan x - \ln(1 + x^2)$

Question 92. The maximum value of the function $y = e^{\frac{x^2 - 3x}{x + 1}}$ on the interval $[0; 3]$ is

- A. e^2 B. e^3 C. 1 D. e

Question 93. Calculate the sum of all integer values of m in the interval $(-21, 21)$ such that the function $y = -x^3 - 3x^2 + mx + 4$ is decreasing on the interval $(0, +\infty)$.

Question 94. The function $f(x)$ has the derivative $f'(x) = \frac{x^{2024}(x - 5)^{2025}}{(x + 1)^{2023}}$. Find the intervals of monotonicity and the number of critical points of the function $g(x) = f(x^2 + 1)$.

Question 95. Find the asymptotes of the curves

(a) $y = \sqrt[3]{1+x^3}$

(b) $y = \ln(1 + e^{-x})$

(c) $y = \frac{x^3 \operatorname{arccot} x}{1+x^2}$

(d) $\begin{cases} x = 2t - t^2 \\ y = \frac{2016t^2}{1-t^3} \end{cases}$

(e) $\begin{cases} x = t \\ y = t + 2 \arctan t \end{cases}$

Question 96. Sketch the curve of

(a) $y = e^{\frac{1}{x}-x}$

(c) $y = \frac{x^3}{x^2+1}$

(b) $y = \sqrt[3]{x^3 - x^2 - x + 1}$

(d) $y = \frac{x-2}{\sqrt{x^2+1}}$

Question 97. Find all integer values of the parameter m such that the maximum value of the function

$$y = \left| \frac{1}{4}x^4 - \frac{19}{2}x^2 + 30x + m - 20 \right|$$

on the interval $[0, 2]$ does not exceed 20.

Question 98. Given the function $f(x) = x^2 - 4x + 3$. How many integer values of m are there such that the equation

$$f^2(|x|) - (m-6)f(|x|) - m + 5 = 0$$

has exactly 6 distinct solutions?

A. 9

B. 11

C. 13

D. 15

Question 99. Given the function $y = f(x)$ with the sign chart of f' provided below, how many critical points does the function $g(x) = f(x^2 - 2x)$ have?

x	$-\infty$	-2	1	3	∞
$f'(x)$		- 0	+ 0	+ 0	-

A. 1

B. 2

C. 3

D. 4

Question 100. How many asymptotes does the graph of the function $y = \frac{1}{2^x - 4}$ have?

A. 2

B. 3

C. 4

D. 5

Question 101. Find the asymptotes of the graph of the function.

(a) $y = \sqrt{\frac{x^3}{x-1}}$

(c) $\begin{cases} x = \frac{t}{t^2-1} \\ y = \frac{t^2}{t-1} \end{cases}$ where t is a parameter.

(b) $y = \frac{x}{\sqrt[3]{x^6-1}}$

(d) $y = \frac{x^{1+x}}{(1+x)^x}$

Question 102. Write the equation of the tangent line at the point $M(1, 1)$ for the curve given by the following parametric equation:

$$\begin{cases} x = t + \cos t \\ y = t^5 + t + 1. \end{cases}$$

Question 103. Prove that $\operatorname{arccot}\left(\frac{1}{x}\right) = \begin{cases} \arctan x & \text{if } x > 0 \\ \pi + \arctan x & \text{if } x < 0. \end{cases}$

Question 104. Find the intersections of the curves $r = \cos 2\varphi$ and $r = \frac{1}{2}$. Sketch the graphs of these functions.

1.7 Week 7

Question 105. Calculate the integrals.

(a) $\int e^{\sin^2 x} \sin 2x dx$

(g) $\int \sin 5x \cos 3x dx$

(m) $\int \left(1 - \frac{1}{x^2}\right) \sqrt{x\sqrt{x}} dx$

(b) $\int (x+2) \ln x dx$

(h) $\int \tan^3 x dx$

(n) $\int \frac{dx}{x\sqrt{x^2+1}}$

(c) $\int |x^2 - 3x + 2| dx$

(i) $\int \frac{dx}{3 \sin x - 4 \cos x}$

(o) $\int \frac{x dx}{(x^2 - 1)^{3/2}}$

(d) $\int \frac{x dx}{(x+2)(x+5)}$

(j) $\int \frac{(3-2x) dx}{\sqrt{1-x^2}}$

(p) $\int \sin x \sin(x+y) dx$

(e) $\int \frac{(x^2+2) dx}{x^3+1}$

(k) $\int \frac{dx}{1 + \sqrt{1+x^2}}$

(q) $\int \frac{1 + \sin x}{\sin^2 x} dx.$

(f) $\int \frac{dx}{(x+a)^2(x+b)^2}$

(l) $\int \frac{(x+1) dx}{\sqrt{x^2-2x-1}}$

1.8 Week 8

Question 106. Calculate the integrals.

(a) $\int \arctan x dx$

(f) $\int \frac{dx}{(x^2 + 2x + 5)^2}$

(b) $\int \frac{x+2}{\sqrt{x^2-5x+6}} dx$

(g) $\int \sin^{n-1} x \sin[(n+1)x] dx, n \in \mathbb{N}^*$

(c) $\int \frac{x dx}{\sqrt{x^2+x+2}}$

(h) $\int e^{-2x} \cos 3x dx$

(d) $\int \frac{x^4 dx}{x^{10}-1}$

(i) $\int \arcsin^2 x dx$

(e) $\int x \sqrt{-x^2+3x-2} dx$

(j) $\int \frac{x^4+1}{x^2+1} dx$

(k) $\int \frac{dx}{x^4 + 1}$

Question 107. Formulate the recurrence relation to calculate I_n , $n \in \mathbb{N}$.

(a) $I_n = \int x^n e^x dx$

(b) $I_n = \int \sin^n x dx$

(c) $I_n = \int \frac{dx}{\cos^n x}$

1.9 Week 9

Question 108. Express the following limit as a definite integral.

$$\lim_{n \rightarrow \infty} \frac{1}{n+1} \left(\frac{1}{\sqrt{4n^2+1}} + \frac{2}{\sqrt{4n^2+2^2}} + \cdots + \frac{n-1}{\sqrt{4n^2+(n-1)^2}} \right)$$

Question 109. Use the definition of the definite integral to find the limits.

(a) $\lim_{n \rightarrow \infty} \left[\frac{1}{n\alpha} + \frac{1}{n\alpha + \beta} + \frac{1}{n\alpha + 2\beta} + \cdots + \frac{1}{n\alpha + (n-1)\beta} \right], (\alpha, \beta > 0)$

(b) $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{1 + \frac{1}{n}} + \sqrt{1 + \frac{2}{n}} + \cdots + \sqrt{1 + \frac{n}{n}} \right)$

Question 110. Given differentiable functions $u(x)$ and $v(y)$, calculate the derivatives.

(a) $\frac{d}{dx} \int_x^{u(x)} e^{t^2} dt$

(b) $\frac{d}{dy} \int_{v(y)}^y e^{t^2} dt$

(c) $\frac{d}{dx} \int_{x^2}^{x^3} \frac{dt}{\sqrt{1+t^4}}$

Question 111. Calculate the limits.

(a) $\lim_{x \rightarrow 0^+} \frac{\int_0^{\sin x} \sqrt{\tan t} dt}{\int_0^{\tan x} \sqrt{\sin t} dt}$

(b) $\lim_{x \rightarrow +\infty} \frac{\int_0^x (\arctan t)^2 dt}{\sqrt{x^2 + 1}}$

(c) $\lim_{x \rightarrow +\infty} \frac{\left(\int_0^x e^{t^2} dt \right)^2}{\int_0^x e^{2t^2} dt}$

Question 112. Calculate the following integrals.

(a) $\int_{1/e}^e |\ln x| (x+1) dx$

(c) $\int_0^{3\pi/2} \frac{dx}{2 + \cos x}$

(e) $\int_0^3 \arcsin \sqrt{\frac{x}{1+x}} dx$

(b) $\int_1^e (x \ln x)^2 dx$

(d) $\int_0^1 \frac{\sin^2 x \cos x}{(1 + \tan^2 x)^2} dx$

(f) $\int_0^{\pi/2} \cos^n x \cos nx dx, n \in \mathbb{N}^*$

Question 113. Prove that if $f(x)$ is continuous on $[0, 1]$, then

$$(a) \int_0^{\pi/2} f(\sin x) dx = \int_0^{\pi/2} f(\cos x) dx$$

$$(b) \int_0^{\pi} x f(\sin x) dx = \int_0^{\pi} \frac{\pi}{2} f(\sin x) dx$$

Apply to calculate the following integrals.

$$(i) \int_0^{\pi} \frac{x \sin x dx}{1 + \cos^2 x}$$

$$(ii) \int_0^{\pi/2} \frac{\sqrt{\sin x} dx}{\sqrt{\sin x} + \sqrt{\cos x}}$$

$$(iii) \int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{\sqrt{2}}}$$

Question 114. Given $a, b > 0$. Compute $\int_{-a}^a \frac{dx}{(1 + b^x)(1 + x^2)}$.

A. $b \arctan a$

C. $\frac{\arctan a}{b}$

E. $2 \arctan a$

B. $a \arctan b$

D. $\arctan a$

F. $\frac{\arctan b}{a}$

Question 115. Given the function $f(x)$ which is continuous on $\mathbb{R}$ and satisfies $f(x^5 + x^3 + 1) = x^2 + x + 1$, calculate $\int_1^3 f(x) dx$.

A. $\frac{420}{2057}$

B. $\frac{2057}{420}$

C. $\frac{2037}{420}$

D. $\frac{420}{2027}$

Question 116. Given the function $y = f(x)$ that satisfies $x = 3^{f(x)} + f(x)$ for all $x \in \mathbb{R}$, calculate $\int_1^4 f(x) dx$.

A. $\frac{7}{2} - \frac{2}{\ln 3}$

B. $\frac{5}{2} - \frac{2}{\ln 3}$

C. $\frac{3}{2} - \frac{2}{\ln 3}$

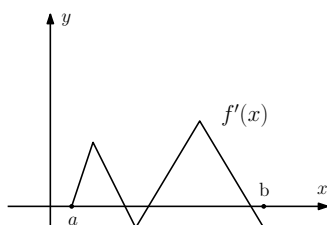
D. $\frac{1}{2} - \frac{\ln 3}{2}$

Question 117. Calculate the following integrals.

$$(a) \int_0^{2\pi} \ln(\sin x + \sqrt{1 + \sin^2 x}) dx.$$

$$(b) \int_{-1}^1 \frac{dx}{1 + x + x^2 + \sqrt{x^4 + 3x^2 + 1}}.$$

Question 118 (Y). Given a differentiable function $y = f(x)$, and the graph of its derivative is provided below. How many distinct real solutions does the equation $f(x) = f(a)$ have in the interval $[a, b]$?



A. 0

B. 1

C. 2

D. 3

Question 119. Prove that $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ -1 & \text{if } x \notin \mathbb{Q} \end{cases}$ is not integrable.

Question 120. Given the function $f(x) = \int_{-x}^{2x} 2^{t^2} dt$, the value of $f'(1)$ is

A. 8

B. 10

C. 34

D. 48

1.10 Week 10

Question 121. Examine the convergence and compute (in the case of convergence) the following integrals.

(a) $\int_{-\infty}^0 x e^x dx$

(d) $\int_0^{+\infty} \frac{dx}{x^2 + 3x + 2}$

(b) $\int_{-\infty}^{+\infty} \frac{dx}{(x^2 + 1)^2}$

(e) $\int_0^1 \frac{\ln x}{\sqrt{x}} dx$

(c) $\int_0^1 \frac{dx}{\sqrt{x(1-x)}}$

(f) $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + (\tan x)^{\sqrt{2}}}$

(g) $(\Upsilon) \int_0^{+\infty} \frac{\arctan(\pi x) - \arctan x}{x} dx$

Question 122. Find the values of a for which the integral converges and evaluate the integral for those values of a .

a) $\int_0^1 \frac{dx}{x^a}$ b) $\int_0^\pi \frac{dx}{\sin^a x}$ c) $\int_1^\infty (\sqrt{\sqrt{x+a} - \sqrt{x}} - \sqrt{\sqrt{x} - \sqrt{x-1}}) dx$

Question 123. Examine the convergence of the following integrals.

(a) $\int_1^{+\infty} \frac{\ln(1+x) dx}{x^2}$

(d) $\int_0^1 \frac{dx}{\tan x - x}$

(g) $\int_0^{+\infty} \frac{\ln(1+3x)}{x\sqrt{x}} dx$

(j) $\int_0^{+\infty} \frac{\sin 2x}{x} dx$

(b) $\int_1^{+\infty} \frac{dx}{\sqrt{x+x^3}}$

(e) $\int_0^1 \frac{\sqrt{x} dx}{\sqrt{1-x^4}}$

(h) $\int_0^{+\infty} \frac{x - \sin x}{\sqrt{x^7}} dx$

(c) $\int_2^{+\infty} \frac{x dx}{\ln^3 x}$

(f) $\int_0^\pi \frac{dx}{\sqrt[3]{\sin x}}$

(i) $\int_0^{+\infty} \frac{\arctan x dx}{\sqrt{x^3}}$

Question 124. If $\int_0^{+\infty} f(x) dx$ converges, can we deduce that $f(x) \rightarrow 0$ as $x \rightarrow +\infty$? Consider the example $\int_0^{+\infty} \sin(x^2) dx$.

Question 125. Given that $f(x)$ is continuous on $[a, +\infty)$ and $\lim_{x \rightarrow +\infty} f(x) = A \neq 0$, does the integral $\int_a^{+\infty} f(x) dx$ converge?

Question 126. (CK20221) Find the real number $m > 0$ such that the following integral converges:

$$\int_0^{+\infty} \frac{dx}{x^m + x^{2m}}.$$

Question 127. (CK20221) Examine the convergence or divergence. Calculate the integral (if it converges).

$$\int_1^{+\infty} \frac{2 \arctan x}{x^2} dx.$$

Question 128. Prove that $\int_0^\infty \frac{dx}{(1+x^2)(1+x^\alpha)}$ doesn't depend on α .

Question 129. Given f is positive, continuous function on $[0, \infty)$ and suppose that f is continuously differentiable on $(0, +\infty)$ and $f'(x) \geq 0$ for all $x > 0$.

(a) Prove that $\int_0^\infty \frac{f'(x)}{f^\alpha(x)} dx$ is convergent for all $\alpha > 1$.

(b) For $\alpha > \frac{1}{2}$, prove that both

$$\int_0^\infty \frac{dt}{f^\alpha(t)} \quad \text{and} \quad \int_0^\infty \frac{dt}{f^\alpha(t) + f'(t)}$$

are convergent or both divergent.

1.11 Week 12

Question 130. Calculate the area of the region bounded by:

(a) The parabola $y = x^2 + 4$ and the line $x - y + 4 = 0$

(b) The curve $y = x^3$ and the lines $y = x, y = 4x, (x \geq 0)$

(c) The circle $x^2 + y^2 = 2x$ and the parabola $y^2 = x, (y^2 \leq x)$

(d) The curve $y^2 = x^2 - x^4$

Question 131. The area of the region bounded by the closed curve $r = 4 \sin \varphi$ is:

A. 4π

B. 2π

C. 3π

D. 8π

Question 132. Find the area of the region bounded by the following curves:

a) $r = \sqrt{3} \cos \varphi, r = \sin \varphi$

c) $r^2 = \sin 2\varphi, r^2 = \cos 2\varphi$

b) $r = 1 + \cos \varphi, r = 1 - \cos \varphi$

d) $r = a \sin \varphi, r = b \cos \varphi, a > 0, b > 0$

Question 133. Calculate the volume of the solid which is the intersection of two cylinders $x^2 + y^2 \leq a^2$ and $y^2 + z^2 \leq a^2, (a > 0)$.

Question 134. Find the volume of the solid bounded by the surface $z = 4 - y^2$, the coordinate planes $x = 0, z = 0$, and the plane $x = a$ ($a \neq 0$).

Question 135. Calculate the volume of the solid of revolution formed by rotating the region bounded by the curves $y = 2x - x^2$ and $y = 0$:

(a) around the x -axis

(b) around the y -axis

Question 136. Calculate the length of the curve:

(a) $y = \ln \left(\frac{e^x + 1}{e^x - 1} \right)$ as x varies from 1 to 2

(b) $\begin{cases} x = a \left(\cos t + \ln \tan \frac{t}{2} \right) \\ y = a \sin t \end{cases}$ as t varies from $\frac{\pi}{3}$ to $\frac{\pi}{2}$, ($a > 0$)

Question 137. Let S be the region bounded by the curves $y = \cos x$ and $y = 0$ for $x \in \left[\frac{3\pi}{2}, \frac{5\pi}{2} \right]$. The formula for the volume of the solid generated by rotating the region S around the y -axis is:

A. $V = \pi \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \cos^2 x \, dx$

C. $V = 2\pi \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \cos^2 x \, dx$

B. $V = 2\pi \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} x \cos x \, dx$

D. $V = \pi \int_0^1 (\arccos x)^2 \, dx$

Question 138. Let S be the region bounded by the curves $y = \cos x$ and $y = 0$ for $x \in \left[\frac{3\pi}{2}, \frac{5\pi}{2} \right]$. The formula for the volume of the solid generated by rotating the region S around the line $x = \frac{\pi}{2}$ is:

A. $V = \pi \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \cos^2 \left(x - \frac{\pi}{2} \right) \, dx$

C. $V = 2\pi \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \cos^2 \left(x - \frac{\pi}{2} \right) \, dx$

B. $V = 2\pi \int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \left(x - \frac{\pi}{2} \right) \cos x \, dx$

D. $V = \pi \int_0^1 (\arccos(x - \frac{\pi}{2}))^2 \, dx$

Question 139. Let S be the region bounded by the curves $y = x^2$ and $y = 3$. Let V be the volume of the solid of revolution generated by rotating the region S around the line $y = 3$. Which of the following is true?

A. $V < 60$

C. $V \geq 75$

B. $60 \leq V < 70$

D. $70 < V < 75$

Question 140. Calculate the surface area of the solid formed by rotating the following curves:

(a) $y = \sin x, 0 \leq x \leq \frac{\pi}{2}$ rotated around the x -axis

(b) $y = \frac{1}{3}(1-x)^3, 0 \leq x \leq 1$ rotated around the x -axis

Question 141. The surface area of the solid generated by rotating the graph of the function $y = x^3$ for $0 \leq x \leq 2$ around the x -axis is $2\pi(a\sqrt{145} + b)$, where $a, b \in \mathbb{Q}$. Find $a + b$:

A. $\frac{17}{54}$

B. $\frac{1}{2}$

C. $\frac{8}{3}$

D. $\frac{1}{3}$

Question 142. Prove that the surface area of the solid generated by rotating the curve $r = r(\varphi)$, for $\alpha \leq \varphi \leq \beta$, where $r(\varphi)$ is a continuously differentiable function and $0 \leq \alpha < \beta \leq \pi$, around the x -axis is:

$$S = \int_{\alpha}^{\beta} 2\pi r(\varphi) |\sin \varphi| \sqrt{(r(\varphi))^2 + (r'(\varphi))^2} \, d\varphi$$

Question 143. Find the formula for the surface area of the solid generated by rotating the curve $r = r(\varphi)$, where $r(\varphi)$ is a continuously differentiable function, for $\alpha \leq \varphi \leq \beta$ around the line $\varphi = \frac{\pi}{2}$.

Question 144. Find the surface area of the solid generated by rotating the curve $r^2 = \cos 2\varphi$ around the line $\varphi = \frac{\pi}{2}$.

Question 145. The curve $y = \sqrt{4 - x^2}$, $-1 \leq x \leq 1$, is a part of the circle $x^2 + y^2 = 4$. Find the surface area of the solid generated by rotating this curve around the x -axis.

Question 146. The parabola $y = x^2$ from $(1, 1)$ to $(2, 4)$ is rotated around the y -axis. Find the surface area of the obtained solid.

Question 147. For $a \in \mathbb{R}$, let $V(a)$ denote the area of the revolution surface of revolution generated by rotating the curve $y = \sqrt{x}$, $0 \leq x \leq 1$, about the horizontal line $y = a$. Find an explicit formula for $V(a)$.

1.12 Week 13

Question 148. Find the domain of the following functions:

(a) $z = \frac{1}{\sqrt{x^2 + y^2 - 1}}$ (c) $z = \arcsin \frac{y-1}{x}$

(b) $z = \sqrt{(x^2 + y^2 - 1)(4 - x^2 - y^2)}$ (d) $z = \sqrt{x \sin y}$

Question 149. The range of the function $z = \operatorname{arccot} \sqrt{3 - x^2 - y^2}$ is

A. $[-\sqrt{3}, \sqrt{3}]$ B. $[\frac{\pi}{3}, \frac{\pi}{2}]$ C. $[\frac{\pi}{6}, \frac{\pi}{2}]$ D. $[0, \sqrt{3}]$

Question 150. Find the limits (if it exists) of the following functions:

(a) $f(x, y) = \frac{y^4}{x^4 + y^2}$, $(x \rightarrow 0, y \rightarrow 0)$ (c) $f(x, y) = \frac{1 - \cos \sqrt{x^2 + y^2}}{x^2 + y^2}$, $(x \rightarrow 0, y \rightarrow 0)$

(b) $f(x, y) = \frac{y^2}{x^2 + 3xy}$, $(x \rightarrow \infty, y \rightarrow \infty)$ (d) $f(x, y) = \frac{x(e^y - 1) - y(e^x - 1)}{x^2 + y^2}$, $(x \rightarrow 0, y \rightarrow 0)$

Question 151. Let $I = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + 2y^4}$. Which of the following statements is true?

A. $I = 1$ C. $I = -1$
B. $I = 0$ D. The limit does not exist.

Question 152. Calculate

(a) $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2)^{x^2 y^2}$ (c) $\lim_{(x,y) \rightarrow (+\infty, +\infty)} \frac{x + y}{x^2 - xy + y^2}$

(b) $\lim_{(x,y) \rightarrow (+\infty, +\infty)} (x^2 + y^2)e^{-(x+y)}$ (d) $\lim_{(x,y) \rightarrow (+\infty, +\infty)} \frac{x^2 + y^2}{x^2 \ln y + y^2 \ln x}$

(a) $A = \sqrt{(2, 02)^3 + e^{0,03}}$

(b) $B = (1, 02)^{1,01}$

Question 164. Prove that if $f(x, y)$ is differentiable at (a, b) , then f is continuous at (a, b) .

Question 165. Apply the chain rule to find the partial derivatives of the following functions:

a) $z = x^4 + x^2y$, $x = s \cos t$, $y = s \sin t$

c) $z = \tan(\frac{u}{v})$, $u = 2s + 3t$, $v = 3s - 2t$

b) $z = \arcsin(x - y)$, $x = s^2 + t^2$, $y = 1 - 2st$

d) $P = \sqrt{u^2 + v^2 + w^2}$, $u = xe^y$, $v = ye^x$, $w = e^{xy}$.

Question 166. Compute $\frac{dy}{dx}$

a) $\arctan(x^2y) = x + xy^2$

b) $e^y \sin x = x + xy$

Question 167. Compute $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$

a) $e^z = xyz$

b) $yz + x \ln y = z^2$

Question 168. Suppose the equation $F(x, y, z) = 0$ defines three variables x, y, z as functions: $z = f(x, y)$, $y = g(x, z)$, and $x = h(y, z)$. If F is differentiable and F'_x , F'_y , and F'_z are non-zero, prove that

$$\frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} = -1.$$

Question 169. Suppose the following functions are all differentiable:

a) If $z = f(x, y)$, where $x = r \cos \varphi$ and $y = r \sin \varphi$, prove that

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \varphi}\right)^2.$$

b) If $z = f(x - y)$, prove that

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0.$$

c) If $z = f(x, y)$, where $x = s + t$ and $y = s - t$, prove that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \frac{\partial z}{\partial t}.$$

Question 170. Let $f(x)$ be continuous on $\mathbb{R}$ and differentiable at 0. Consider the function

$$F(x, y) = \begin{cases} \frac{1}{x} \int_x^{xy} f(t) dt & \text{if } x \neq 0, \\ (y - 1)f(0) & \text{if } x = 0. \end{cases}$$

a) Find the first partial derivatives F'_x , F'_y .

b) Prove that these first partial derivatives are continuous on $\mathbb{R}^2$.

Question 171. Let $\varphi(x)$ be a continuously differentiable function twice on $\mathbb{R}$, $\varphi(0) = 0$, $\varphi''(0) \neq 0$, and the two-variables function $f(x, y)$ on $\mathbb{R}^2$ defined by

$$f(x, y) = \begin{cases} \frac{x\varphi(y) - y\varphi(x)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Prove that f is continuous on $\mathbb{R}^2$ but its first partial derivatives are not continuous on $\mathbb{R}^2$.

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Question 172. Assume $z = yf(x^2 - y^2)$, where f is a differentiable function. Prove that for the function z , the following identity always holds:

$$\frac{1}{x}z'_x + \frac{1}{y}z'_y = \frac{z}{y^2}.$$

Question 173. Find the partial derivatives of the following composite functions:

(a) $z = e^{u^2-2v^2}, u = \cos x, v = \sqrt{x^2 + y^2}$

(c) $z = \arcsin(x - y), x = 3t, y = 4t^3$

(b) $z = \ln(u^2 + v^2), u = xy, v = \frac{x}{y}$

Question 174. Let f be a differentiable function twice on $\mathbb{R}$. Prove that the function $w(x, t) = f(x - 3t)$ satisfies the wave equation:

$$\frac{\partial^2 w}{\partial t^2} = 9 \frac{\partial^2 w}{\partial x^2}.$$

Question 175. Find the derivatives and partial derivatives of the following implicit functions:

(a) $x^3y - y^3x = a^4$, find y'

(c) $\arctan \frac{x+y}{a} = \frac{y}{a}$, find y'

(b) $x^2 + y + z^3 + e^z = 0$, find z'_x, z'_y

(d) $x^3 + y^3 + z^3 - 3xyz = 0$, find z'_x, z'_y

Question 176. Given the implicit function $z = z(x, y)$ defined by the equation $2x^2y + 4y^2 + x^2z + z^3 = 3$, calculate $\frac{\partial z}{\partial x}(0, 1), \frac{\partial z}{\partial y}(0, 1)$.

Question 177. Given $u = \frac{x+z}{y+z}$, calculate u'_x, u'_y , knowing that z is an implicit function of x, y defined by the equation $ze^z = xe^x + ye^y$.

Question 178. The equation $z^2 + \frac{z}{x} = \sqrt{y^2 - z^2}$ defines an implicit function $z = z(x, y)$. Prove that

$$x^2z'_x + \frac{1}{y}z'_y = \frac{1}{z}.$$

Question 179. Find the second-order partial derivatives of the following functions:

(a) $z = \frac{1}{3}\sqrt{(x^2 + y^2)^3}$

(b) $z = x^2 \ln(x + y)$

(c) $z = \arctan \frac{y}{x}$

(d) $z = \sin(x^3 + y^2)$

Question 180. Find the second differential of the following functions:

(a) $z = xy^3 - x^2y$

(b) $z = e^{2x}(x + y^2)$

(c) $z = \ln(x^3 + y^2)$

Question 181. Given $z = x^2 + 2xy + 3y^2 + 3x + 4y + 5$. Calculate d^2z .

A. $d^2z = 2(dx)^2 + 2dxdy + 4(dy)^2$

D. $d^2z = 2(dx)^2 + 4dxdy + 4(dy)^2$

B. $d^2z = 2(dx)^2 + 2dxdy + 3(dy)^2$

C. $d^2z = (dx)^2 + 4dxdy + 4(dy)^2 + 3dx$

E. $d^2z = 2(dx)^2 + 4dxdy + 6(dy)^2$

Question 182. Given the function

$$f(x, y) = \begin{cases} \frac{x^3 - 4y^3}{x^2 + 2y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Calculate $\frac{\partial f}{\partial x}(0, 0) + \frac{\partial f}{\partial y}(0, 0)$.

- A. 0 B. -1 C. -2 D. -3 E. -4 F. -5

Question 183. Calculate the value of the expression $A = xz'_x + yz'_y$ with $z = -\ln r$ and $r = \sqrt{x^2 + y^2}$.

- A. -2 B. -1 C. 0 D. 1 E. 2

Question 184. The implicit function $z(x, y)$ is defined by the equation $2x^2y + 4y^2 + x^2z + z^3 = 7$. Calculate $z'_y(0, \sqrt{2})$.

- A. $\frac{7\sqrt{3}}{2}$ B. $-\frac{7\sqrt{2}}{3}$ C. $\frac{7\sqrt{2}}{3}$ D. $-\frac{8\sqrt{2}}{3}$ E. $\frac{8\sqrt{2}}{3}$

Question 185. Given $z = y^{2x}$. Calculate $d^2z(1, 1)$.

- A. $4dxdy - 2dy^2$ B. $2dxdy + 2dy^2$ C. $4dxdy + 2dy^2$ D. $2dx^2 + 4dxdy$ E. $2dx^2 - 4dxdy$

Question 186. Prove that $f''_{xy}(0, 0) \neq f''_{yx}(0, 0)$, where

$$f(x, y) = \begin{cases} x^2 \arctan \frac{y}{x} - y^2 \arctan \frac{x}{y} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Question 187. If $f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2} & \text{if } x^2 + y^2 \neq 0 \\ 0 & \text{if } (x, y) = (0, 0), \end{cases}$ then does $f''_{xy}(0, 0)$ exist?

Question 188. Given that f is a function differentiable up to the second order. Calculate dz and d^2z , where

- a) $z = f(\sqrt{x^2 + y^2})$ b) $z = f(u, v)$, with $u = x + y$ and $v = x - y$.

Question 189. Calculate f''_{xy} and f''_{yx} , where $f(x, y) = \sin(xy)$.

Question 190. Prove that $f''_{xy}(0, 0) \neq f''_{yx}(0, 0)$, where

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Question 191. Calculate f'''_{xyz} if $f(x, y, z) = \sin(3x + yz)$.

Question 192. Given $z = \frac{1}{3}\sqrt{(x^2 + y^2)^3}$. Calculate d^2z and d^3z at $(3, 4)$.

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Question 193. Find the Maclaurin polynomial (Taylor polynomial at the origin) for the following functions:

a) $f(x, y) = \ln(1 + x + y)$

b) $f(x, y) = e^x \sin y$

Question 194. Find the Taylor polynomial up to the second degree at the point $A(1, 1)$ for the function $f(x, y) = x^y$.

Question 195. Find the shortest distance from the point $(3, 4, 0)$ to the cone $z = \sqrt{x^2 + y^2}$.

Question 196. Find the critical points of the following functions:

(a) $z = 4x^3 + 6x^2 - 4xy - y^2 - 8x + 2$ (c) $z = 4xy - x^4 - 2y^2$ (e) $z = e^{2x}(4x^2 - 2xy + y^2)$

(b) $z = 2x^2 + 3y^2 - e^{-(x^2 + y^2)}$ (d) $z = \frac{4}{x} + \frac{3}{y} - \frac{xy}{12}$ (f) $z = x^3 + y^3 - (x + y)^2$

Question 197. Find the extreme values of the function $f(x, y) = x^2 + y^2 + xy - 5x - 4y + 10$ with the constraint $x + y = 4$.

Question 198. Find the extreme values of the function $f(x, y) = 6 - 4x - 3y$ with the constraint $x^2 + y^2 = 1$.

Question 199. Find the extreme values of the function $f(x, y, z) = x + y + z^2$ with the constraints

$$\begin{cases} z - x = 1 \\ y - xz = 1 \end{cases}$$

Question 200. Find the maximum and minimum values of the function $f(x, y) = x^2 - y^2$ on the domain $D : x^2 + y^2 \leq 1$.

Question 201. A rectangular box without a bottom is made from 12m^2 of cardboard. Find the maximum volume of the box.

Question 202. Use Lagrange multipliers to find the maximum and minimum values of the following functions with the given constraints:

(a) $f(x, y, z) = xyz; \quad x^2 + 2y^2 + 3z^2 = 6$ (b) $f(x, y, z) = x^4 + y^4 + z^4; \quad x^2 + y^2 + z^2 = 1$

Question 203. Find the maximum and minimum values of the volume of a rectangular box, given that its surface area is 1500 cm^2 and the total length of its edges is 200 cm .

Question 204. Find the points on the curve $xy^2z^3 = 2$ that are closest to the origin.

Question 205. Find the critical points of the function $z = x^2 + y^2$ with the constraint $3x - 4y = 5$.

Question 206. Find a point on the ellipse $4x^2 + y^2 = 4$ that is farthest from the point $A(1, 0)$.

Question 207. Find the maximum and minimum values of the following functions:

(a) $z = x^2 + y^2 + xy - 7x - 8y$ in the triangle bounded by the lines $x = 0$, $y = 0$, and $x + y = 6$

(b) $z = 4x^2 - 9y^2$ in the region bounded by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

Question 208. The minimum value of the function $z = x^2 - 3xy + 4y^2 + x - 5y$ is:

- A. -2 B. -1 C. 0 D. -3 E. does not exist

Question 209. Let the maximum and minimum values of the function $f(x, y) = x^2 + y^2 - 4x$ on the rectangle $[-1, 1] \times [-2, 2]$ be M and m , respectively. Then $M + m$ is:

- A. 4 B. 6 C. 8 D. 2